

JK FLIP FLOP

JK Flip-Flop – Inputs and Outputs					
Signal	Type	Size	MSB	LSB	
J	Input	1-bit	—	—	
K	Input	1-bit	—	—	
clk	Input	1-bit	—	—	
reset	Input	1-bit	—	—	
Q	Output	1-bit	—	—	

MAIN CODE

```
module jk_flipflop (  
    input wire J,  
    input wire K,  
    input wire clk,  
    input wire reset,  
    output reg Q  
);  
  
always @(posedge clk or posedge reset) begin  
    if (reset)  
        Q <= 1'b0;  
    else begin  
        case ({J,K})  
            2'b00: Q <= Q;    // Hold  
            2'b01: Q <= 1'b0; // Reset  
            2'b10: Q <= 1'b1; // Set  
            2'b11: Q <= ~Q;   // Toggle  
        endcase  
    end  
end
```

```
        end
    end

endmodule

TEST BENCH CODE

`timescale 1ns/1ps

module jk_flipflop_tb;

    reg J, K, clk, reset;
    wire Q;

    jk_flipflop uut (
        .J(J),
        .K(K),
        .clk(clk),
        .reset(reset),
        .Q(Q)
    );

    always #5 clk = ~clk;

    initial begin

        clk = 0;
        reset = 1;
        J = 0;
        K = 0;
```

```
#10 reset = 0;
```

```
// Hold
```

```
J = 0; K = 0; #10;
```

```
// Set
```

```
J = 1; K = 0; #10;
```

```
// Reset
```

```
J = 0; K = 1; #10;
```

```
// Toggle
```

```
J = 1; K = 1; #10;
```

```
// Toggle again
```

```
J = 1; K = 1; #10;
```

```
$finish;
```

```
end
```

```
initial begin
```

```
    $monitor("Time=%0t | J=%b K=%b | Reset=%b | Q=%b",
```

```
        $time, J, K, reset, Q);
```

```
end
```

```
endmodule
```